human

Show me the building where OSID = '502c37bc-408b-4156-ac74-8d07946cffbe in Exeter

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': "User wants to find and show the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' in Exeter. Provide the general steps to solve this geospatial query."}

planning_agent

The query specifies a unique identifier (OSID) for a building, which indicates that we are looking for a specific entity. The geographical area is Exeter, so we will focus on that location. There is no ambiguity in the query as it clearly states the OSID and the area.

output steps: ["Find Exeter 1 area", "Search for building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' in Exeter 1 result"]

host_agent

Tool send_message returned result : The query specifies a unique identifier (OSID) for a building, which indicates that we are looking for a specific entity. The geographical area is Exeter, so we will focus on that location. There is no ambiguity in the query as it clearly states the OSID and the area.

output steps: ["Find Exeter 1 area", "Search for building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' in Exeter 1 result"]

Calling tool generate_metadata_for_all_artifacts with args : {} Tool generate_metadata_for_all_artifacts returned result : {} Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.,\n count of records fetched : 98.\n ',] Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']} Tool generate_metadata_for_artifacts returned result : [metadata summary for 'exeter_area_polygon']

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns for high quality
    # Check for 'Exeter' in name1_text, and also look for relevant description/descriptiongroup
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter'))) &
                   (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and settlement-related categories."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon corresponding specifically to Exeter, filtered using name1_text and relevant settlement descrip
    return [summary, artifact_name, artifact_description, filtered]

filter_exeter_area_polygon(data)
```

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns for high quality
    # Check for 'Exeter' in name1_text, and also look for relevant description/descriptiongroup
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter', na=False))) &
                   (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and settlement-related categories."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon corresponding specifically to Exeter, filtered using name1_text and relevant settlement descrip
    return [summary, artifact_name, artifact_description, filtered]
```

Output: ['Filtered to 1 polygon(s) for Exeter using name and settlement-related categories.',] Output from agent data_analysis_agent : Filtered to 1 polygon for Exeter using the name and settlement-related categories. The result is saved as 'filtered_exeter_area_polygon', which contains the polygon(s) from 'exeter_area_polygon' specifically corresponding to Exeter, filtered using the 'name1_text' and relevant 'descriptiongroup' columns. Tool send_message returned result : ["Filtered to 1 polygon for Exeter using the name and settlement-related categories. The result is saved as 'filtered_exeter_area_polygon', which contains the polygon(s) from 'exeter_area_polygon' specifically corresponding to Exeter, filtered using the 'name1_text' and relevant 'descriptiongroup' columns.Addtionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['Polygon(s) from exeter_area_polygon corresponding specifically to Exeter, filtered using name1_text and relevant settlement descriptiongroup.']", []]

named_area_agent

Output from agent named_area : The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This artifact contains the polygon(s) corresponding specifically to Exeter. Tool send_message returned result : ["The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This artifact contains the polygon(s) corresponding specifically to Exeter.Addtionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['Polygon(s) from exeter_area_polygon corresponding specifically to Exeter, filtered using name1_text and relevant settlement descriptiongroup.']", []]

host_agent

Calling tool send_message with args : {'target': 'buildings', 'task_description': "Search for the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within the 'filtered_exeter_area_polygon' artifact."}

buildings_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': None, 'street_address': None, 'filename': 'osid_search_in_filtered_exeter_area'} Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : [\'buildings_osid_search_in_filtered_exeter_area\', \'buildingline_osid_search_in_filtered_exeter_area\', \'buildingpart_osid_search_in_filtered_exeter_area\'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within the 'filtered_exeter_area_polygon' artifact. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Search for the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within the 'filtered_exeter_area_polygon' artifact. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' within the 'filtered_exeter_area_polygon' artifact. using filters None."],\n counts of records fetched are : [69169, 69342, 85480].\n ', [, ,]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'buildings_osid_search_in_filtered_exeter_area' to find the building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe'. Save the filtered result as 'building_with_osid_502c37bc_filtered_exeter_area'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_osid_search_in_filtered_exeter_area']} Tool generate_metadata_for_artifacts returned result : [metadata summary for 'buildings_osid_search_in_filtered_exeter_area']

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_building_by_osid(data: list):
    gdf = data[0]
    filtered = gdf[gdf['osid'] == '502c37bc-408b-4156-ac74-8d07946cffbe']
    summary = f"Filtered to {len(filtered)} building(s) with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe'."
    artifact_name = 'building_with_osid_502c37bc_filtered_exeter_area'
    artifact_description = "GeoDataFrame containing the building in the filtered Exeter area with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe'. All columns re
    return [summary, artifact_name, artifact_description, filtered]
```

Output: The search return no results and no artifacts are generated

Output from agent data_analysis_agent : No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found in the provided filtered Exeter area dataset. No results were returned for this search. Tool send_message returned result : No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found in the provided filtered Exeter area dataset. No results were returned for this search.

buildings_agent

Output from agent buildings : No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found within the 'filtered_exeter_area_polygon' dataset. There are no results for this search. If you need to search in a different area or have another OSID to check, please let me know. Tool send_message returned result : No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found within the 'filtered_exeter_area_polygon' dataset. There are no results for this search. If you need to search in a different area or have another OSID to check, please let me know.

host_agent

Output from agent host_agent : No building with OSID = '502c37bc-408b-4156-ac74-8d07946cffbe' was found within the Exeter area.

If you would like to: - Check a different OSID, - Search in a different area, - Or need help with another building or location,

please let me know how you'd like to proceed!